

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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RELATED SUPPORT - [Programme of Activity requirements](#)

This document contains the following Sections

Key Project Information

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KEY PROJECT INFORMATION

Type of VPA	☐ Real case VPA ☒ Regular VPA
Scale of VPA Note that a VPA can be of one scale. Please select applicable scale accordingly.	☐ Microscale ☒ Small scale ☐ Large scale
Title of corresponding real case VPA (if applicable)	Not applicable
GS ID of real case VPA (if applicable)	Not applicable
GS ID of VPA	GS11564
Title of VPA	GS1385 Man and Man Enterprise Improved Cooking Stoves Programme in Ghana - VPA002 (Western region)
Time of First Submission Date	05/04/2022
Date of Design Certification	14/03/2023
Version number of the VPA-DD	2.2
Completion date of version	11/07/2023
Coordinating/managing entity	Man and Man Enterprise
VPA Implementer (s)	Man and Man Enterprise
Project Participants and any communities involved	Man and Man Enterprise
Host Country (ies)	Ghana
GS ID and Title of applicable Design Certified VPA	Not applicable
GS ID and Title of applicable Performance Certified VPA	Not applicable
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities

	☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Other Requirements applied	--
Methodology (ies) applied and version number	AMS II.G Version 12.0 "Energy efficiency measures in thermal applications of non-renewable biomass"
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
VPA Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory) 13.2 Integrate climate change measures into national policies, strategies and planning	Emission reduction Indicator : Amount of GHGs emissions avoided or sequestered	191,627	tCO _{2e}
SDG: 1 No Poverty 1.1 By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day	1.1.1 Proportion of the population living below the international poverty line by sex, age, employment status and geographical location (urban/peri urban/rural) Indicator : Average household savings i.e., decrease in expenditure on basic service such cooking, lighting, drinking	43.26	USD
SDG:3 Good Health and Well Being 3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	3.9.1 - Mortality rate attributed to household and ambient air pollution Indicator : Household perception of health benefits (reduction in the incidence of eye or respiratory diseases)	100	%
7 Affordable and Clean Energy 7.1 By 2030, ensure universal access to affordable, reliable and modern energy services	7.1.2 Proportion of population with primary reliance on clean fuels and technology Indicator : Number of beneficiaries : households	45,288	Number

8 Decent Work and Economic Growth

8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value

8.5.1 Average hourly earnings of female and male employees, by occupation, age and persons with disabilities

Indicator: Total number of jobs created by the project

10 Number

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

The proposed Gold Standard SSC-VPA “GS1385 Man and Man Enterprise Improved Cooking Stoves Programme in Ghana - VPA002 (Western region)”, hereafter referred to as “VPA002” is carried out within the urban, peri urban and rural areas of Western region, Ghana, where households mainly rely on charcoal for cooking purposes with inefficient devices (coal pots). An average of 0.180 t¹ of woody biomass is used per person (for cooking purposes) annually. The promotion and dissemination of over 240,000 affordable and efficient improved cookstoves (ICS) to low-income Ghanaian households and the associated awareness and training campaigns will help Ghanaian households by replacing currently used traditional coal pot², thus reducing Greenhouse Gas emissions by 191,627 tCO₂/yr.

Man and Man Enterprise is the coordinating/managing entity (CME) and also act as the VPA implementer and the technology supplier for the proposed VPA.

Ghanaian households will have the opportunity to buy ICS directly from Man and Man Enterprise and/or through vendors at an affordable price by entering into an agreement with Man and Man Enterprise (by signing a sales agreement), in which they commit to participate durably, ease the monitoring work necessary for project success, relinquish the ownership of the carbon credits generated by their cook stove to the VPA implementer and benefit from specific replacement conditions.

The VPA under consideration is not claiming for other carbon credit certification schemes other than Gold Standard under the Voluntary market.

A.1.1. Eligibility of the VPA under approved PoA

1 From FAO value <https://www.fao.org/3/i6935e/i6935e.pdf>

2 According to (Hedon, 2012), coal pot cooking stove have an efficiency range from 15 to 18%. A conservative value of 18% is taken.

Table 2 Eligibility for VPA inclusion as per PoA requirements

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
1	Project Area, Project Boundary and Scale /Location of project	The SSC-VPA takes place within the borders of Ghana (a host country which has ratified the Kyoto protocol and is listed as a Non-Annex I country with no cap GHG emissions); Only Carbon Dioxide (CO₂), Methane (CH₄) and/or Nitrous Oxide (N₂O) are eligible for Gold Standard crediting, provided Projects comply with Gold Standard Requirements and eligibility criteria.	Mapped locations of stoves distribution area provided The SSC-VPA boundaries are all the centers (urban, peri urban, rural) of Western region in Ghana which are within borders of Ghana such as demonstrated by the GPS coordinates in section A.2. ³ Cross-check with registries of other standards revealed SSC-VPA is not included in other standards. Methodology AMS II.G V.12.0 applied through the PoA-DD is limited to reductions in CO₂. Ghana does not have an emission cap in place at the moment of validation of this VPA-DD.
2	Specification of the technology/measure	The SSC-VPA specifies the level and type of service provided by the technology/measure as well as its performance which are in line with the technology outlined in SSC-PoA-DD and in compliance with national and/or international testing/certification requirements. Specifications of the technology/measure shall include the type, capacity and other key features of the design of the systems;	Certified ICS testing report of medium size ICS is provided. The SSC-VPA involves the marketing of improved cook stoves, with a certified efficiency of 30.9% which is at least 20% such as described in section A.3. ICS deployed are Jiko type system technology composed of two parts, a metallic part and a ceramic part that retain the heat and make it more efficient than traditional system. The capacity (power output of the system is 0.95 KW). ICS deployed are portable.
3	Target group	The target group of each SSC-VPA shall be either: - The households (Households are domestic unit consisting of the members of a family who live together and share	Section A.1. Project description in VPA-DD And Project Design study and/or any references that can substantiate the compliance.

³ The sale agreement of each ICS sold under the VPA will be collected with information about address in order to check that each ICS is sold to users in Ghana but cannot be provided at the time of VPA inclusion thus it will be reflected in the monitoring report.

	food and cooking equipment). - The communities (communities are groups of individuals that have an identity and share some of their resources such as biomass and cooking equipment. For instance, schools, police camps, prisons are considered as communities) - The small and medium enterprises (Small and Medium Enterprises (SME) are small and medium business that use stoves as equipment for their business and consume biomass in order to sell meals (commercial use). For instance, restaurant is considered as a SME as they use ICSs to cook and sell meals.). - Charcoal stove users and/or three-stone fire users.	/ The SSC-VPA involves the marketing of ICS for charcoal and using households of urban, peri urban and rural centers of Ghana as described in section A.3.
4 Compliance with the applicability of the applied methodologies	The baseline consumption and baseline stove efficiency of target group selected should be determined.	AMS.II-G version 12.0 FAO. 2017. The charcoal transition: greening the charcoal value chain to mitigate climate change and improve local livelihoods, by J. van Dam. Rome, Food and Agriculture Organization of the United Nations The SSC-VPA involves the marketing of ICS for charcoal and using households of urban, peri urban and rural centers of Ghana as described in section A.5. / The SSC-VPA involves the marketing of ICS for <i>households</i>, which are consuming in the baseline 5.64 tonnes of wood-fuel equivalent per year <i>in the absence of the project activity</i>.⁴

⁴ Baseline consumption may be updated along project implementation if more recent data become available.

		The efficiency of baseline stoves is determined at 20%
5	Compliance with the applicability of the applied methodologies / Type of project The project shall be the distribution (sale) of efficient cook stoves to target user in Ghana, and therefore classifies as an End-use Energy Efficiency Improvement project.	Section A.5. Project description in VPA-DD / The SSC-VPA involves the sales of ICS such as stated in section A.5 and thus demonstrates that it classifies as an End-use Energy Efficiency Improvement project. Since there are GS approved CSA requirements, the project type is automatically eligible (cf. para. 4.1.3 of GS4GG Principles and requirements.) End-use energy efficiency such as efficient cooking is a pre-identified CSA project type, as per para. 3.1.1 of GS4GG CSA requirements V.1.2.
6	Demonstration of additionality In reference to the "Annex B (Positive List) of the CSA Requirements, V. 1.2", the annual energy savings generated by each ICS sold under the framework of a VPA included under this PoA shall not exceed 600 MWh, which is equivalent to 1,800 MWh_{th} (microscale CDM unit);	Detailed calculations are provided in Section A.4.: Application of methodology(ies) in VPA-DD And ICS test report. / The SSC-VPA nominal energy savings of each ICS Nominal energy saving of each ICS <i>medium size</i> is 8.62 MWh_{th}/y which is lower than 600 MWh, which is equivalent to 1,800 MWh_{th}
7	Compliance with the applicability of the applied methodologies. The VPA complies and applies the small-scale baseline and monitoring methodology AMS II.G Version 12.0;	Section B.2.: Application of methodology(ies) in VPA-DD. And The SSC-VPA properly applies methodology AMS-II.G Version 12.0 in compliance with its eligibility criteria as follow: AMS II.G version 12.0 comprises the introduction of improved cookstoves which is typically the case of the proposed SSC-VPA. It is proven that non-renewable biomass has been used and forest area has reduced since 1990⁵. .

⁵ The World Bank. 2020. Forest area (sq. km) – Ghana table (<https://data.worldbank.org/indicator/AG.LND.FRST.K2?locations=GH>)

8 Other requirements of the applied methodologies	Transfer of credits ownership shall be guaranteed from stove user to VPA implementer. The VPA is either implemented by the Coordinating/managing entity or by another entity which has signed a binding agreement with the CME which ensures that they are aware and agree that their activity is subject to a PoA and that their carbon rights have to be relinquished to the CME;	Contractual agreement between CME and VPA implementer (necessary only if CME is not the VPA implementer). And / or Sales agreement between VPA implementer and end user. / The SSC-VPA is implemented by Man and Man Enterprise which is also the CME of the PoA thus no contractual agreement is needed. Each end user included into the program voluntarily adheres to its terms and conditions, including a waiver for the transfer of credit ownership to the project owner, Man and Man Enterprise, either through individual sales agreements or retailers' charter.
9 Double counting	Each cook stove sold under the SSC-VPA will be uniquely marked using ICS serial number and recorded in an electronic database for all PoA life, as described in SSC-PoA-DD section B in order to ensure that no double counting will occur ;	Section C of the SSC-PoA and template of sale contract in which serial number of ICS that will be distributed under the PoA appears and show serial number structure ensuring that ICS to be allocated with a unique serial no to ensure no double counting will occur. / As stated in section C of PoA-DD and section A.13 of VPA-DD, each ICS sold under the SSC-VPA will be identified based on the unique serial number riveted under the stove and recorded in order to avoid double counting intra PoA (misallocation or double counting within the different VPAs of the PoA) and double counting inter-PoA (misallocation between two different PoAs or related to cook stoves not belonging to the PoA).
10 Unregistered/deregistered activity	The SSC-VPA-DD has been reviewed by the CME and submitted to a VVB for inclusion into the PoA; Among others, the CME shall check that VPAs are neither registered as project activities, included in another registered PoAs, nor the project activities that have been deregistered;	Conclusion review of the SSC-VPA-DD of the CME. And/or Quotes of VVB signed for VPA inclusion. / A letter from the CME to the VVB (cc/ VPA implementer) submitting the proposed VPA-DD for inclusion into the PoA is proving that the SSC-VPA has been verified by CME and submitted to a VVB for review before inclusion in the PoA.

11	Local stakeholder consultation and environmental impact analysis	A local stakeholder consultation has been conducted prior to the inclusion of the SSC-VPA in the PoA The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	Section F of PoA-DD or section E. of the VPA-DD, LSC report submission notification and consultation minutes and presentation. / A local stakeholder consultation has been conducted at the VPA level for household's end-users in urban areas. Details pertaining to the consultation are available in LSC report.
12	Sampling	The SSC-VPA sampling plan complies with the "<i>Standard for sampling and surveys for CDM project activities and programme of activities</i>" V.08.0;	Section B.7.2 of SSC-VPA-DD and / or additional supporting evidence if applicable. / As stated in the monitoring plan outlined in the PoA-DD, the sampling effort aims to achieve for the annual inspection the level of accuracy (95/10 if cross-VPA monitoring is chosen or 90/10 if standalone VPA monitoring is chosen) which is compliant with AMS-II.G. version 12.0 and the "<i>Standard for sampling and surveys for CDM project activities and programme of activities</i>".
13	Diversion of official development assistance	The SSC-VPA does not involve any public funding according to the OECD definitions for Official Development Assistance (ODA);	ODA Declaration (as per GHG Emissions Reduction & Sequestration Product Requirements) / As stated in A.7, the project does not involve any public funding according to the OECD definitions for Official Development Assistance (ODA).
14	Start date	The starting date of the SSC-VPA is verifiable through documentary evidence and shall not be prior the start date of the PoA.	Document evidence such as order of first batch of ICSs or first sale agreement signed. / As stated in section C.1.1, the starting date of the SSC-VPA has been determined as the sale of the first ICS which is verifiable the first sale agreement signed by household which should not be before the start date of PoA.
15	Gold Standard requirements (SDG and Safeguarding Principles assessments)	The SSC-VPA shall respect the requirements of the Gold Standard regarding SDG and Safeguarding Principles assessments.	VPA-DD and related evidence / The VPA describes in section the Local Stakeholder Consultation Report and sections B and D of the VPA-DD how it will respect the requirements of the Gold Standard regarding SDG and Safeguarding

		Principles assessments, as well as monitoring of SDG parameters.
16	Local stakeholder consultation and environmental impact analysis	An Environmental Impact Assessment (EIA) of the project shall be conducted if required by the national laws. A local consultation took place in Tarkwa on the 9th December, 2021, in order to share information with the local residents/entities of interest, and obtain feedback from them. More details available on the SCR. / Since no Environmental Impact Assessment (EIA) is required by the national laws for VPA002, no EIA report is provided.
17	Gold Standard requirements (Host Country Requirements)	Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations. This project is in compliance with Ghana's legal, environmental, and social regulations. The project is actively contributing to the Host Country by reducing the GHG emissions, fuel consumption as well as economic and time savings to thousands of poor households. The company is registered and has legal right to operate. Therefore no other permission/permit is required. M&M also received report from EPA regarding environmental impact of stoves production.
18	Gold Standard requirements (Contact details)	The Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). GS Cover letter Extract from the trade registries and documentation by the governing jurisdiction / Cf. Appendix 1 of VPA-DD and GS cover letters provided. Extract from the trade registries and documentation by the governing jurisdiction shall be provided.
19	Gold Standard Requirements (Legal Ownership)	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Sample of sale agreements / Cf. section A.3 of VPA-DD

The proofs that end-users are aware of and willing to give up their rights on Products shall be provided.

A.1.2. Legal ownership of products generated by the VPA and legal rights to alter use of resources required to service the project

>> The project owner, Man and Man Enterprise, has full and uncontested legal ownership of the carbon credits that are generated under the Gold Standard Certification. Each end user included into the program voluntarily adheres to its terms and conditions, including a waiver for the transfer of credit ownership to the project owner, Man and Man Enterprise, either through individual sales agreements or retailers' charter.

A.2. Location of VPA

The VPA002 will be implemented in all urban, peri-urban and rural centers of the Western region in Ghana.

Range of GPS coordinates of Ghana are taken as a reference for the project boundary. The sales agreement of each ICS sold under the VPA will be collected with information about the address in order to check that each ICS included is in Ghana. This will be reflected in the monitoring report. Moreover, each cook stove will be uniquely identified during manufacturing phase.

Table 2 – Location of the project activity implemented under the VPA

Boundaries	GPS coordinates
Ghana	4.7115 – 11.1700 / -3.2685 – 1.2080 (decimal units)



Figure 1 – Map of Ghana and the Western Region in red

A.3. Technologies and/or measures

Compared to the currently used traditional coal pot, the ICS to be included in the VPA002 is an improved cook stove allowing quicker heating-up, longer cooking and heat retaining with less woodfuel as well as lower combustion fumes from Jiko-type technology⁶. Improved cook-stoves (ICS) are a mature technology for the efficient energy conversion of biomass to heat. The technology that will be disseminated within this VPA is the so called Jiko-type improved cookstove. The ICS is made of two different parts: the metallic body and the ceramic liner. Its performance lies in the production of the liner made of three different type of clay which retain the heat released by the combustion of the fuel and avoid losses. The adoption of this innovative technology results in significant savings of woodfuel and associated expenses.

The improved cook stove has been designed and manufactured by Man and Man Enterprise. Beyond improved emissions and efficiency, Man and Man Enterprise's stoves are durable, affordable, and safe (Figure 2 below). ICS to be deployed under VPA002 is the following type:

⁶ Jiko type system improved cooking stove is an improved cooking stove technology from Kenya. The specific design feature of this technology is the fact that stoves are composed of two parts, a metallic part and a ceramic part that retain the heat and make it more efficient than traditional system.



Figure 2 – Preview of Man and Man Enterprise’s improved cookstoves be disseminated for households

Table 3 – Medium size (for households) project’s cook stoves technical specifications

<i>Weight</i>	Approx. 7 kg
<i>Cooking power output</i>	0.95 kW _{th} ⁷
<i>Efficiency</i>	Approx. 30.9% (new)
<i>Type of fuel</i>	Charcoal
<i>Adoption</i>	Traditional cooking styles and posture
<i>Durability</i>	5 years ⁸
<i>Hight</i>	Approx. 25 cm
<i>Width</i>	Approx. 30.5 cm

The technology and know-how being applied by the project activity is environmentally safe and sound since it will positively curb deforestation currently endangering natural forests by promoting sane and sustainable practices in the targeted areas of Ghana. It is made possible by highly efficient stoves technology transfer to the Host party.

The VPA002 is focused on the replacement of traditional stoves with very low efficiencies (locally known as “coal-pots”) commonly used by charcoal-using Ghanaian households, with an improved stove known as the “Holy cook”. There are four different sizes of improved cook stove produced by the project proponent however only the medium size for household usage will be disseminated within this VPA. The

⁷ Results from Water Boiling Tests done by the Lomé University laboratory are available to the VVB

⁸ The technical lifetime of the ICS is 5 years. Lifetime of ICS is derived from manufacturer’s experience and the monitoring data from the VPA1 included in the PoA GS1385

ICS that will be sold to households and probably in major quantity either through Man and Man Enterprise and/or vendors within the PoA has an average thermal efficiency of approx. 30.9% (new) and is therefore significantly more efficient than the stoves to be replaced.

Only medium size ICS will be distributed under VPA002. Table below presents the standardized medium size ICS to be disseminated within the VPA for households with its dimensions and typical applications.

Stove size	Typical application
Medium	Commonly used for both domestic (households till 6 persons) and non-domestic (commercial or institutional) applications

Positive contribution to the Sustainable Development Goals

Besides being one of the main causes of deforestation in Africa, the social and environmental benefits of improved cooking stoves are multiple, for example:

- they reduce woman workloads and time involved in fuel collection, dependence on a scarce resource and related expenses by reducing fuel wood domestic consumption significantly;
- by reducing indoor air pollution, they can reduce the risk of respiratory diseases, especially for women and children. Exposure to smoke from cooking contributes to a range of chronic illnesses and acute health impacts such as early childhood pneumonia, emphysema, cataracts, lung cancer, bronchitis, cardiovascular disease, and low birth weight. Women and young children are the most affected, with more than 2,200 children in Ghana dying every year as a result of acute lower respiratory infections caused by the use of solid fuels;

Moreover, PoA activities will foster employment for local communities (marketing campaigns, sales network) throughout its lifecycle.

Overall, the activity contributes to the following sustainable development goals:

SDG 1 End poverty in all its forms everywhere

SDG 3 Ensure healthy lives and promote well-being for all at all ages

SDG 7 Ensure access to affordable, reliable, sustainable and modern energy for all

SDG 8 Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

SDG 13 Take urgent action to combat climate change and its impacts

A.4. Scale of the VPA

The VPA is a small-scale activity, as per para. 3.1.2 c) of the CSA requirements V.1.2⁹, which covers “project activities that improve thermal energy efficiency, the maximum energy saving of 60 GWh(e) per year is equivalent to 180 GWh_(th) per year saving’ (para. 126 (b) of CDM project standard for programmes of activities V.2).

The proposed SSC-VPA appliance will qualify as a micro-scale CDM unit as it applies UNFCCC methodology and will remain under the limits of 600 MWh (which is <20 GWh_{th}) of annual energy savings per appliance (=CDM microscale unit, as per para. 12 of the Methodological tool: Demonstration of additionality of microscale project activities, V. 9.0 in combination with para. 14 & 15 (cf. eligibility criteria in section A.2).

Considering efficiency of the baseline stove to be 20% and the one of ICS to be deployed of 30.9%, therefore energy savings per stove per year are equal to: $(1 - (20\% / 30.9\%)) * B_{old,HH} * NCV = 35.27\% * 5.64 \text{ tonnes /y} * (15.6 \text{ GJ/tonne} * 1\text{h}/3.6) = 8.62 \text{ MWh}_{th}/\text{y}$ which is less than the microscale threshold which amounts to 20 GWh_{th}/y. It is also less than 600 MWh, equivalent to 1,800 MW_{th}/y, representing the threshold of energy savings per unit of project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions. Such activities are deemed additional, as per para. 4.1.9 and Annex B (Positive List) of the CSA Requirements, V. 1.2.

A.5. Funding sources of VPA

The PoA is financed by private funding, thanks to VER sales to private, voluntary offset buyers. Also, it gains revenue via the sale of ICS to end-users, at a 50% subsidy price per unit.

⁹ Para. 3.1.2 (c) states, “for the purpose of applying UNFCCC methodologies for quantification of GHG reductions, ‘small scale’ is defined as in CDM Modalities and Procedures for three projects types;”

The VPA does not expect to involve any public funding according to the OECD definitions¹⁰ for Official Development Assistance (ODA). A written declaration of project's non-use of ODA has been submitted to VVB using the ODA declaration template form.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

The chosen methodology is the approved baseline and monitoring small-scale methodology AMS-II.G "Energy efficiency measures in thermal applications of non-renewable biomass" (Version 12.0).

In line with the application of AMS-II.G methodology, the VPA refers to the following documents:

- General guidelines to SSC CDM methodology (Version 23.0)
- TOOL21: Demonstration of additionality of small-scale project activities (Version 13.1)
- TOOL19: Demonstration of additionality of microscale project activities (Version 9.0)

The methodology also refers to the latest approved versions of the following approved standards, methodology(ies) and tool(s):

- (a) "AMS-III.BG.: Emission reduction through sustainable charcoal production and consumption" (Version 3.0)
- (b) "TOOL30: Calculation of the fraction of non-renewable biomass" (Version 2.0)
- (c) "Standard: Sampling and surveys for CDM project activities and programme of activities" (Version 8.0)

B.2. Applicability of methodology (ies)

Methodology AMS-II.G V.12, section 6.3, stipulates that "the use of this methodology in a project activity under a program of activities is legitimate if the following leakages

¹⁰ Based on: OECD, Development Assistance Committee, Glossary, available online at <https://www.oecd.org/dac/dac-glossary.htm#ODA>

are estimated and accounted for, as required on a sample basis using a 90/30 precision for the selection of samples:

- (a) Use of non-renewable woody biomass saved under the project activity to justify the baseline of other CDM project activities can also be a potential source of leakage. If this leakage assessment quantifies a portion of non-renewable woody biomass saved under the project activity that is then used as the baseline of other CDM project activities then $B_{old,i,j}$ is adjusted to account for the quantified leakage;
- (b) Increase in the use of non-renewable woody biomass outside the project boundary to create non-renewable woody biomass baselines can also be a potential source of leakage. If this leakage assessment quantifies an increase in the use of non-renewable woody biomass outside the project boundary then $B_{old,i,j}$ is adjusted to account for the quantified leakage;
- (c) As an alternative to subparagraphs (a) and (b), $B_{old,i,j}$ can be multiplied by a net to gross adjustment factor of 0.95¹¹ to account for both leakages, in which case surveys are not required.

Conclusion: As alternative (c) is chosen and applied in this small-scale VPA and all other conditions of the methodology are integrated into the PoA-DD/VPA-DD or met through the eligibility criteria, the small-scale VPA is eligible to be included in the PoA.

In the literature, the distribution of improved cookstoves in Ghana and its positive impacts on the environment and health of users has been established since 1996¹². More recently, studies confirmed the capacity of standards like the CDM and Gold Standard to help the financing of ICS distribution in the country¹³.

B.3. VPA boundary

According to methodology AMS-II.G, the spatial extent of the project boundary is the physical, geographical site of the efficient devices that utilize biomass.

¹¹ Paragraph 39 and paragraph 48 of the methodology allow the use of a net to gross adjustment factor of 0.95 in lieu of conducting a survey to account for leakage emissions. In the case of a VPA opting to apply the adjustment factor, the adjustment factor is only applied once, i.e. either paragraph 39 or paragraph 48 (c) is applied. Also, the adjustment factor does not need to be applied twice for option (a) and (b).

¹² Adam, Iddrisu, "Energy and deforestation in Ghana: A study of woodfuel-deforestation links in rural Ghana" (1996). Theses and Dissertations (Comprehensive). 345.

¹³ Ribeiro, Joseph & Owusu, Emmanuel & Ayesu, Edmund & Antwi, Edward & Ahiekpor, Julius & Bensah, Edem. (2015). Adoption and Utilization of Improved Cookstoves in Ghana. International Journal of Innovative Research and Development. 4. 305-311.

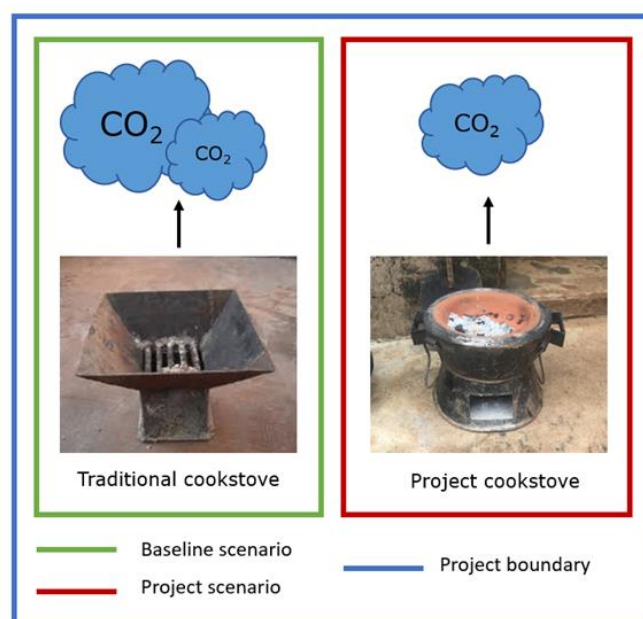


Figure 3: Delineation of project stove from baseline stove

Source		GHGs	Included?	Justification/Explanation
Baseline	CO ₂ emissions from consumption of non-renewable woody biomass in low-efficiency three-stone fires and traditional cook stoves	CO ₂	Yes	Main emission source
		CH ₄	No	Minor emission source (neglected for simplification)
		N ₂ O	No	Minor emission source (neglected for simplification)
Project	CO ₂ emissions from consumption of non-renewable woody biomass in improved cook stoves distributed by the project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Minor emission source (neglected for simplification)
		N ₂ O	No	Minor emission source (neglected for simplification)

B.4. Establishment and description of baseline scenario

Baseline technology:

Approximately 67% of households in Ghana use polluting fuels as their main cooking fuels together with traditional cookstoves. These are mainly firewood (33.3%) and charcoal (34.1%). Firewood is predominant in the rural areas (63%), while charcoal (44.3%) and LPG (34.8%) dominate in the urban areas as the main cooking fuels. Also,

62.5% of Ghana's household¹⁴ use traditional coalpot or three stone fire which are inefficient stoves.

Inefficient coal pots are notoriously wasteful; other drawbacks of traditional biomass stoves include the diffusion of heat during windy conditions, the difficulty of controlling the fire, users' exposure to heat and smoke as well as fire hazards. The baseline scenario is a continuation of the current situation i.e. the use of non-renewable woodfuel in inefficient traditional cook stove (coal pots) by households throughout Ghana.

The concept of suppressed demand is not applied.

B.5. Demonstration of additionality

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	At time of first project design certification, deemed additionality has been proven by applying para. 9 (b) and para. 17 of version 07.1 of the CDM "Tool for microscale additionality": "9. Energy efficiency project activities that aim to achieve energy savings at a scale of no more than 20 gigawatt hours per year are additional if any one of the conditions below is satisfied: (a) The geographic location of the project activity is in an LDC/SIDS or SUZ of the host country identified by the government in accordance with the paragraph 8(a)(i) above; (b) The project activity is an energy efficiency activity with both conditions (i) and (ii) below satisfied: (i) Each of the independent subsystems/measures in the project activity achieves an estimated annual energy savings equal to or smaller than 600 megawatt hours; (ii) End users of the subsystems or measures are households/communities/ SMEs. 17. For CPAs applying microscale thresholds at the unit level rather than at the aggregate level of the CPA, the term 'project activities' in paragraphs 8 - 12 and 14 above shall be read as 'units'18. If each of the units contained in the CPA satisfies the condition to qualify as a 'microscale CDM unit', then the coordinating/managing entity is not required to demonstrate compliance of the CPA with the microscale or small-scale thresholds at the aggregate level of the CPA. In such cases, the requirements related to debundling stated in paragraphs 13 and 16 above do not apply either." As per CSA requirements V.1.2:
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¹⁴ The landscape of energy for cooking in Ghana: A review, Modern Energy Cooking Services, 2021. url : <https://mecs.org.uk/wp-content/uploads/2021/02/The-landscape-of-energy-for-cooking-in-Ghana-A-review.pdf>

	<i>"4.1.8 All projects seeking the issuance of Certified Impact Statements and/or Products shall demonstrate Financial Additionality in accordance with the Principles & Requirements and the applicable Product requirements.</i> <i>4.1.9 Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification:</i> <i>(a) Positive list (Annex B of this document)</i> <i>(b) Projects located in LDC, SIDS, LLDC4</i> <i>(c) Microscale projects"</i> As per para. 1.1.3 of Annex B of the CSA requirements V.1.2, the following project activities form part of the positive list: <i>"Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year."</i> 600 MWh translates to 1,800 MWh_{th}.
Describe how the proposed project meets the criteria for deemed additionality.	The proposed PoA applies microscale thresholds at the unit level, meeting para. 16 and both conditions of para. 8 (b) of the microscale tool: The proposed PoA is solely composed of isolated units (Improved Cookstoves) where the users of the technology are households, communities or small and medium enterprises, each appliance resulting in less than 600 MWh_{th} of annual energy savings. Each unit results in about 8.62 MWh_{th} of energy savings per year, which is ≤ 600 MWh or 1,800 MWh_{th}. The conditions are part of the eligibility criteria and are checked at the time of VPA inclusion. The project thus meets the conditions for deemed additionality.

B.5.1. Prior Consideration

Project start date : 12/07/2021

Stakeholder consultation (1st round) date : 09/12/2021

Date of first submission to Preliminary Review : 05/04/2022

As the 1st round of the stakeholder consultation was conducted after the project start date, the project is retroactive.

However, revenues from carbon credit were already acknowledge and taken into consideration for this project as it takes part into a Programme of Activities (PoA) with

consistent Project Participants (Man & Man Enterprise), project design and business model. This GS1385 Man and Man Enterprise Improved Cooking Stoves Programme in Ghana PoA has a start date on 25/03/2013, and its renewed crediting period spans from 02/09/2020 to 01/09/2025.

B.5.2. Ongoing Financial Need

N/A¹⁵

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
SDG 1 No poverty	1.1	Average household savings i.e., decrease in expenditure on basic service such cooking, lighting, drinking
SDG 3 Good health and well being	3.9	Household perception of health benefits (reduction in the incidence of eye or respiratory diseases)
SDG 7 Affordable clean energy	7.1	Number of beneficiaries: Households
SDG 8 Decent work	8.5	Total number of jobs created
SDG 13 Climate action	13.2, 13.b	Amount of GHGs emissions avoided or sequestered

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

¹⁵As per template guide (07/10/2020), "this information need only be included at Design Certification Renewal and only for those projects that are required to demonstrate financial additionality." As shown further above, this project is not required to demonstrate financial additionality since included in the positive list of Annex B of the CSA requirements V.1.2.

SDG 1: No Poverty

Applied methodology/approach	Equation/calculation
Proportion of the population living below the international poverty line by sex, age, employment status and geographical location (urban/rural) Approach: 1. Estimate ex-ante the decrease in expenditure on cooking fuel, thanks to the project device use.	Net Benefit (SDG 1) = $HH_{Baseline}$ Where: $HH_{Baseline}$ Cost of woody biomass that is saved by the project device use

SDG 3: Good Health and Well Being

Applied methodology/approach	Equation/calculation
3.9.1 Mortality rate attributed to household and ambient air pollution Approach: Monitoring Surveys conducted to % users reporting reduction in smoke/PM after shifting to ICS in project	Ex-post Monitoring Surveys Records Net Benefit (SDG 3) = $SPM_{Project} - SPM_{Baseline}$ Where: $SPM_{HH,Baseline}$ % HH reporting reduction in smoke/PM emissions while cooking on improved stove in baseline $SPM_{HH,Project}$ % HH reporting reduction in smoke/PM emissions while cooking on improved stove in project

SDG 7: Affordable and Clean Energy

Applied methodology/approach	Equation/calculation
7.1.2 Proportion of population with primary reliance on clean fuels and technology Approach: Record the number of stoves distribute and monitor the proportion of operational stoves	ICS distribution records + Ex-post Monitoring Surveys Records Net Benefit (SDG 7) = $N_{o,i,j} * P_{operational\ stoves,y}$ Where: $N_{o,i,j}$ Number of stoves distributed to households $P_{operational\ stoves,y}$ Percentage of operational stoves

SDG 8: Decent Work and Economic Growth

Applied methodology/approach	Equation/calculation
8.5.1 Average hourly earnings of female and male employees, by occupation, age and persons with disabilities Approach:	Employment records Net Benefit (SDG 8) = $QE\ IG_{Project} - QE\ IG_{Baseline}$ Where: $QE\ IG_{Baseline}$ Quantitative Employment and income generation (Number of person (male and female) hired under Baseline)

Recording the number of employees (male / female) in the project under administrative, sales, production and management positions	QE IG _{Project} Quantitative Employment and income generation (Number of person (male and female) hired under Project)
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SDG 13: Climate Action

Applied methodology/ approach	Equation/calculation
Amount of CO ₂ e emissions reduced by the project per year Approach: AMS-II.G Version 12.0	The overall GHG reductions achieved by the project activity will be calculated as follows: $ER_y = \sum \sum ER_{y,i,j} - LE_y$ Where: $ER_{y,i,j} = By_{savings,i,j} \times No_{i,j} \times ny_{i,j} \times \mu_y \times fNRB_{,y} \times NCVbiomass \times EF_{projected_fossil\ fuel}$ Where: $By_{savings,i,j}$ = Quantity of woody biomass that is saved per cookstove device of type i and batch j during year y (tonnes) $fNRB_{,y}$ = Fraction of woody biomass that can be established as non-renewable biomass (fraction or %) $NCVbiomass$ = Net calorific value of the non-renewable woody biomass that is substituted (IPCC default for wood fuel, 0.0156 TJ/tonne, based on the gross weight of the wood that is 'air-dried') $EF_{projected_fossilfuel}$ = Emission factor of fossil fuels projected to be used to substitute nonrenewable woody biomass by similar consumers (tCO₂e/TJ). $NO_{i,j}$ = Number of project devices of type i and batch j commissioned (number) $ny_{i,j}$ = Proportion of commissioned project devices of type i and batch j ($NO_{i,j}$) that remain operating in year y (fraction) μ_y = Adjustment to account for any continued use of pre-project devices during the year y

B.6.2. Data and parameters fixed ex ante

SDG 13

Data/parameter	B_{old,p}
Unit	tonnes /person/year
Description	Annual quantity of charcoal that would have been used per person in the household in the absence of the project

	activity to generate thermal energy equivalent to that provided by the project devices.
Source of data	FAO. 2017. The charcoal transition: greening the charcoal value chain to mitigate climate change and improve local livelihoods, by J. van Dam. Rome, Food and Agriculture Organization of the United Nations (https://www.fao.org/3/i6935e/i6935e.pdf)
Value(s) applied	0.180
Choice of data or Measurement methods and procedures	AMS-II.G. allows for the use of historical data or survey of local usage to define relevant baseline appliance types as described in the baseline scenario. The VPAs utilize a historical data from FAO regarding average charcoal consumption per capita and per year.
Purpose of data	Calculation of baseline emissions
Additional comment	-

Data/parameter	$N_{p,HH}$
Unit	Number
Description	Average number of persons served per household prior to project implementation
Source of data	Established ex ante prior to project implementation based on records of households served by the project
Value(s) applied	5.22
Choice of data or Measurement methods and procedures	Lowest value from all monitoring periods of 5 Man&Man ICS dissemination VPAs and CPAs, representing a sample of 3000+ surveys. The 5.22 value is conservative with regard to the global mean of 5.89.
Purpose of data	Calculation of baseline emissions
Additional comment	-

Data/parameter	$B_{old,HH}$
Unit	tonnes/household/year
Description	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate thermal energy equivalent of that provided by the project devices
Source of data	This parameter shall be determined ex ante
Value(s) applied	5.64 for households
Choice of data or Measurement methods and procedures	$B_{old,p}$ times $N_{p,HH}$ times $M_{wood/mcharcoal}$: 0.18 t/p/yr times 5.22 p/HH/yr x 6
Purpose of data	Calculation of baseline emissions
Additional comment	-

Data/parameter	$B_{old,i,j}$
Unit	tonnes/year
Description	Annual quantity of woody biomass that would have been used in the absence of the project activity to generate thermal energy equivalent to that provided by the project device type i and batch j
Source of data	$B_{old,HH}$
Value(s) applied	5.64
Choice of data or Measurement methods and procedures	$B_{old,HH}$ divided by $N_{d,HH}$: 5.64 divided by 1
Purpose of data	Baseline emissions
Additional comment	$B_{old,i,j}$ equals $B_{old,HH}$ when only one project device per household is distributed

Data/parameter	$f_{NRB,y}$
Unit	Fraction
Description	Fraction of woody biomass saved by the project activity in year y that can be established as non-renewable biomass.
Source of data	FAO and IPCC data and other sources of information (as per Tool30: Calculation of the fraction of non-renewable biomass, Version 02)
Value(s) applied	0.7520
Choice of data or Measurement methods and procedures	As per para.51 to 52 of AMS-II.G and Tool30: Calculation of the fraction of non-renewable biomass, Version 02.0. Calculation of f_{NRB} , where Tool30 guidelines were used to determine the local value, is detailed in sheet 'fNRB' of the ER ex-ante document.
Purpose of data	Calculation of Baseline emissions
Additional comment	-

Data/parameter	$NCV_{biomass}$
Unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass, briquettes or charcoal used in project devices
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value(s) applied	0.0156
Choice of data or Measurement methods and procedures	IPCC default for wood fuel, 0.0156 TJ/tonne, based on the gross weight of the wood that is 'air-dried' may be used if fuel used in project device is also woody biomass.
Purpose of data	Calculation of baseline emissions

Additional comment	-
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Data/parameter	Leakage_{adj}
Unit	Fraction
Description	Net to gross adjustment factor to account for leakages
Source of data	AMS-II.G V.12.0
Value(s) applied	0.95
Choice of data or Measurement methods and procedures	In case this leakage adjustment factor is applied, it is not required to survey the use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable energy sources.
Purpose of data	Calculation of baseline emissions
Additional comment	B _{old} is multiplied by a net to gross adjustment factor of 0.95 to account for leakages according to AMS-II.G version 12.0

Data/parameter	m_{wood}/m_{charcoal}
Unit	kg biomass/kg charcoal
Description	Conversion factor wood/charcoal
Source of data	IPCC 1996 Reference Manual, Chapter 1, Energy, p 1.45 as noted in AMS II.G. http://www.ipcc-nggip.iges.or.jp/public/gl/guidelin/ch1ref3.pdf
Value(s) applied	6
Choice of data or Measurement methods and procedures	The default value should be applied according recommendation of the IPCC Guidelines if no national/local value is available.

Purpose of data	Calculation of baseline emissions
Additional comment	-

Data/parameter	$\eta_{old,i}$
Unit	Fraction
Description	Efficiency of pre-project device
Source of data	AMS II.G Version 12.0 "Energy efficiency measures in thermal applications of non-renewable biomass"
Value(s) applied	0.20
Choice of data or Measurement methods and procedures	Default efficiency value for pre-project device from AMS II.G Version 12.0
Purpose of data	Calculation of Baseline emissions
Additional comment	-

Data/parameter	EF_{projected_fossilfuel}
Unit	tCO ₂ /TJ
Description	Emission factor for the fossil fuels projected to be used for substitution of non-renewable woody biomass by similar consumers
Source of data	Para. 25 of AMS II.G (CO ₂ and non-CO ₂ GHG emissions)
Value(s) applied	73.2
Choice of data or Measurement methods and procedures	Default regional values of the fossil fuel emission factor: Ghana forms part of Sub-Saharan Africa.
Purpose of data	Calculation of baseline emissions

Additional comment	-
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SDG 1

Data/parameter	HH_{Baseline}
Unit	USD
Description	Cost of woody biomass that is saved by the project device use per year
Source of data	See SDG Impact Tool
Value(s) applied	43.26
Choice of data or Measurement methods and procedures	See SDG Impact Tool
Purpose of data	SDG 1 Impact calculation
Additional comment	-

SDG3

Data/parameter	SPM_{HH,Baseline}
Unit	%
Description	% HH reporting reduction in smoke/PM emissions while cooking on improved stove in baseline
Source of data	-
Value(s) applied	0
Choice of data or Measurement methods and procedures	-
Purpose of data	SDG 3 Impact calculation

Additional comment	-
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SDG8

Data/parameter	QE IG_{Baseline}
Unit	Number
Description	Quantitative Employment and income generation (Number of person (male and female) hired under Baseline)
Source of data	-
Value(s) applied	0
Choice of data or Measurement methods and procedures	-
Purpose of data	SDG 8 Impact calculation
Additional comment	-

B.6.3. Ex ante estimation of SDG Impact

Sustainable Development Goals Targeted	SDG Impact Indicator	Calculation	Ex-ante Estimate
1 No Poverty	Indicator: Average household savings i.e., decrease in expenditure on basic service such cooking, lighting, drinking	$HH_{Baseline}$	43.26 USD per year
3 Good Health and Well Being	Indicator: Household perception of health benefits (reduction in the incidence of eye or respiratory diseases)	$SPM_{Project} - SPM_{Baseline}$	100%
7 Affordable and Clean Energy	Indicator: Number of beneficiaries: Households	$N_{o,i,j} * n_{y,i,j}$	45,288 per year

8 Decent Work and Economic Growth	Indicator: Total number of jobs created	$QE\ IG_{Project} - QE\ IG_{Baseline}$	10
13 Climate Action	Indicator : Amount of CO ₂ e emissions reduced by the project per year	Refer table below	191,627 tCO ₂ e

B.6.4. Summary of ex ante estimates of each SDG outcome

Emission reductions (tCO₂e or VERs) are calculated as:

SDG 13: Climate action (VERs)

Year	Baseline estimate	Project estimate	Leakage	Net benefit
12/07/2021-31/12/2021	20,197	0	0	20,197
2022	102,774			102,774
2023	162,935	0	0	162,935
2024	223,095	0	0	223,095
2025	283,256	0	0	283,256
01/01/2026-11/07/2026	165,878	0	0	165,878
Total	958,135	0	0	958,135
Total number of crediting years	5			
Annual average over the crediting period	191,627	0	0	191,627

SDG 1: No poverty: estimation of end-users money savings (detailed in SDG impact tool)

Year	Baseline estimate	Project estimate	Net benefit
12/07/2021-31/12/2021	14.46	0	14.46
2022	43.26	0	43.26
2023	43.26	0	43.26
2024	43.26	0	43.26
2025	43.26	0	43.26
01/01/2026-11/07/2026	28.80	0	28.80
Total	216.30	0	216.30
Total number of crediting years	5		
Annual average over the crediting period	43.26	0	43.26

SDG 3: Reduced kitchen smoke exposure : proportion of end-users reporting less smoke

Year	Baseline estimate	Project estimate	Net benefit
12/07/2021-31/12/2021	0	100%	100%
2022	0	100%	100%
2023	0	100%	100%
2024	0	100%	100%
2025	0	100%	100%
01/01/2026-11/07/2026	0	100%	100%
Total	0	100%	100%
Total number of crediting years	5		

Annual average over the crediting period	0	100%	100%
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SDG 7: Number of beneficiaries : households

Year	Baseline estimate	Project estimate	Net benefit
12/07/2021-31/12/2021	0	30,600	30,600
2022	0	73,800	73,800
2023	0	117,000	117,000
2024	0	160,200	160,200
2025	0	203,400	203,400
01/01/2026-11/07/2026	0	226,440	226,440
Total	0	226,440	226,440
Total number of crediting years	5		
Annual average over the crediting period	0	45,288	45,288

SDG 8: Number of decent jobs created.

Year	Baseline estimate	Project estimate	Net benefit
12/07/2021-31/12/2021	0	10	10
2022	0	10	10
2023	0	10	10
2024	0	10	10
2025	0	10	10
01/01/2026-11/07/2026	0	10	10

Total	0	10	10
Total number of crediting years	5		
Annual average over the crediting period	0	10	10

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13 & SDG 7

Data / Parameter	N_{o,i,j}
Unit	-
Description	Number of commissioned project devices of type i and batch j
Source of data	Monitoring (CME records - Electronic database used for registering all ICS's sold)
Value(s) applied	Value to be updated ex-post
Measurement methods and procedures	During project activity implementation, the following data shall be recorded: (a) Number of new devices distributed under the project activity, identified by the type of devices and the date of commissioning (b) Data to unambiguously identify the recipient of the new devices distributed under the project activity (e.g., name, address, phone number). New stoves installed during the ongoing monitoring period will claim ERs only from the start date of operation, not from the start date of the monitoring period.
Monitoring frequency	continuous
QA/QC procedures	Databases will be periodically checked by the CME. Each stove directly sold will be traced from manufacturer to user using its unique ID. The count cannot exceed the total distributed number of stoves at any point in time, as recorded in the electronic database.
Purpose of data	Assessment of Gold Standard sustainability indicators.

	Calculation of baseline emissions
Additional comment	After 5 years of lifetime, stoves are automatically and conservatively erased from the database to prevent for their count in ERs, although a significant proportion of them are still in operation.

Data / Parameter	μ_y
Unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y
Source of data	Ex-ante estimation
Value(s) applied	0.8584
Measurement methods and procedures	Measured/Calculated. During the monitoring visit, CME-mandated field agents establish if the other stoves (besides the ICS) are used. Field agents estimate the usage rate of the pre-project stove(s) "by formulating questions and/or collecting evidence to determine the frequency of usage of both the project devices and baseline devices. A sample of target end-users is randomly selected, the participating target end-user visited and asked about their average weekly or daily fuel consumption, to obtain one day of measurement. This parameter is calculated by taking the ratio of meals per day (or week) cooked using the baseline (pre-project)/other stove. The adjustment to account for any continued use of pre-project devices is calculated as follows: $1-\mu_y$ = Number of meals cooked over the other stoves / N meals cooked by the target end-user
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	-
Purpose of data	Assessment of Gold Standard sustainability indicators Calculation of baseline emissions
Additional comment	Monitoring takes place during ordinary time (if there is a festival, funeral, or anything else out of the ordinary planned, the monitoring should be delayed).

Data / Parameter	$\eta_{new,y,i,j}$
Unit	Fraction
Description	Efficiency of the medium size device being deployed as part of the project activity for a specific age class j
Source of data	-
Value(s) applied	30.9 % (cf. Lomé test report 2013)
Measurement methods and procedures	According to AMS.II.G efficiency shall be measured/estimated as per the following: [..] (2) manufacturer specifications on efficiency based on water boiling test (WBT) may be used. The sampling test of stoves by such certification bodies/agents or manufacturers is conducted following a 90/10 precision in accordance with the "Standard for sampling and surveys for CDM project activities and programme of activities." [..] WBT equipment was calibrated in compliance with applicable national or international standards and D.7.2 of VPA-DD.
Monitoring frequency	At each monitoring period (WBT for each batches)
QA/QC procedures	-
Purpose of data	Assessment of Gold Standard sustainability indicators; Baseline Emissions
Additional comment	-

Data / Parameter	$n_{y,i,j}$
Unit	-
Description	Proportion of commissioned project devices of type i and batch j ($N_{0,i,j}$) that remain operating in year y (fraction)

Source of data	Capped value from Technologies and Practices to Displace Decentralized Thermal Energy Consumption ¹⁶ (TPDDTEC) methodology
Value(s) applied	0.90
Measurement methods and procedures	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve the required precision. A discount shall be applied based on the percentage of devices operational as determined by the sample survey, e.g. if survey shows that 10% of the devices is non-operating, an adjustment factor shall be applied to number of project devices commissioned in a particular batch. To meet TPDDTEC usage rate guidelines, the proportion value shall be capped at 0.90. Separate samples shall be taken for each batch. Consequently, $n_{y,i,j}$ is the proportion of cooking stoves found to be operating in a representative sample, i.e. $n_{y,i,j}$ (= stoves operating) or its reverse $1-n_{y,i,j}$ (stoves not operating). The VPA implementer will keep a database of all stoves sold and account for distributed cook stoves that would not be operating or not be found during monitoring, although estimated minimal: Participating users voluntarily acquire improved cook stoves for a fee and are thus not likely not to use it subsequently.
Monitoring frequency	-
QA/QC procedures	Sampling is conducted by applying the 95/10 confidence precision for the sample size calculation (if conducted biennially).
Purpose of data	Assessment of Gold Standard sustainability indicators; Calculation of baseline emissions + SDG 7 contribution
Additional comment	In addition to operating status, information regarding the baseline stove used by target end user will be recorded. If baseline stove is not included under baseline defined under the specific VPA, the new device will be counted as not operating thus no Emission Reduction will be accounted. This disincentives fuel switches from lower carbon to higher carbon fuel stoves and ensures the established baseline remains valid.

¹⁶ See TPDDTEC methodology on this URL : <https://globalgoals.goldstandard.org/407-ee-ics-technologies-and-practices-to-displace-decentralized-thermal-energy-tpddtec-consumption/>

Data / Parameter	Date of commissioning of batch <i>j</i>
Unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch.
Source of data	Internal records
Value(s) applied	Recorded ex-post
Measurement methods and procedures	Every time an ICS is sold a sale agreement is filled and an electronic database is filled.
Monitoring frequency	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
QA/QC procedures	-
Purpose of data	Calculation of baseline emissions
Additional comment	To be reported in the monitoring report

Data / Parameter	Date of commissioning of project device <i>i</i>
Unit	Date
Description	Actual date of commissioning of the project device.
Source of data	Internal records
Value(s) applied	Recorded ex-post
Measurement methods and procedures	Every time an ICS is sold a sale agreement is filled and an electronic database is filled. Based on the information collected in the electronic database, the date of commissioning is determined, assuming conservative lead time between construction/installation and start operation.

	Direct sales lead time : 22 days Resales lead time : 29 days Details on lead time calculation will be available in the ER sheet
Monitoring frequency	Fixed and recorded at the time of commissioning/distribution of project devices
QA/QC procedures	-
Purpose of data	Calculation of baseline emissions
Additional comment	

Data / Parameter	$N_{d,HH}$
Unit	Number
Description	Number of project devices distributed per household
Source of data	Internal records
Value(s) applied	1
Measurement methods and procedures	-
Monitoring frequency	Recorded at the time of commissioning/distribution of project devices
QA/QC procedures	Verified during monitoring campaign.
Purpose of data	Calculation of baseline emissions
Additional comment	If a household purchases more than one cooking stove, the user database will account for any additional project device, which will be reflected in adjustment factor $N_{d,HH}$. Ex-ante estimation of $N_{d,HH} = 1$.

SDG 3

Data / Parameter	$SPM_{HH,project}$
Unit	%

Description	% HH reporting qualitative air improvements from better combustion efficiency and reduced consumption of fuel
Source of data	Ex-post monitoring surveys
Value(s) applied	100 % of respondents to confirm their exposition to less smoke and/or health problems
Measurement methods and procedures	Sum of users confirming to be exposed to less smoke divided by sum of all users.
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	SDG 3 contribution
Additional comment	-

SDG 8

Data / Parameter	QE IG_{project}
Unit	- (number)
Description	Quantitative Employment and income generation. Number of person (male and female)
Source of data	HR records
Value(s) applied	10
Measurement methods and procedures	Assessed qualitatively
Monitoring frequency	Annually
QA/QC procedures	- Invoices of safety equipment purchased - On-site checks by CME - Training certificates
Purpose of data	SDG 8 contribution
Additional comment	On qualitative employment : - All ceramic liner employees are provided with safety boot, over-all attire, and nose mask. - All metal fabrication employees are provided with safety boot, over-all attire, ear muff as well and safety helmet. - Training of production operators and monitoring agents in safety measures

B.7.2. Sampling plan

The project monitoring will comply with:

- GS4GG PROGRAMME OF ACTIVITY REQUIREMENTS, Version 1.2, para. 12.1.10, which states that:

(e) The Monitoring Plan for the activity is designed as per the applied methodology and in accordance with the PoA DD;

- Monitoring methodology AMS-II.G Version 12.0, whereby it is stated that:

47. Relevant parameters shall be monitored and recorded during the crediting period as indicated in section 6.1 below. The applicable requirements specified in the "General guidelines for SSC CDM methodologies" are also an integral part of the monitoring guidelines specified below and therefore shall be followed by the project participants.

48. A statistically valid sample of the locations where the devices are deployed, with consideration, in the sampling design, of occupancy and demographic differences can be used to determine parameter values used to calculate emission reductions, as per the relevant requirements for sampling in the "Standard for sampling and surveys for CDM project activities and programmes of activities". When biennial inspection is chosen a 95 per cent confidence interval and a 10 per cent margin of error shall be achieved for the sampling parameter. On the other hand, when the project proponent chooses to inspect annually, a 90 per cent confidence interval and a 10 per cent margin of error shall be achieved for the sampled parameters. In cases where survey results indicate that 90/10 precision or 95/10 precision are not achieved, the lower bound of the 90 per cent or 95 per cent confidence interval of the parameter value may be chosen as an alternative to repeating the survey efforts to achieve the 90/10 or 95/10 precision.

49. Efficiency of devices may be monitored in a common survey with other monitoring parameters; therefore, a random sub-sample within the common survey can be taken for which stove efficiency is tested, as long as the required precision for stove efficiency is achieved.

- Standard for "Sampling and surveys for CDM Project Activities and Programme of Activities" V.9, paragraph 22 – Sampling requirements for PoAs:

"Parameter values shall be estimated by sampling in accordance with the requirements in the applied methodology separately and independently for each of the CPAs included in a PoA except when a single sampling plan covering a group of CPAs is undertaken applying 95/10 confidence/precision for the sample size calculation."

and para. 14:

"If the sample size calculation returns a value of less than 30 samples, a minimum sample size of 30 shall be chosen when the parameter of interest is a proportion. If the parameter of interest is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be used if the resulting sample size is less than 30."

- General Guidelines for SSC-CDM methodologies, whereby it is stated in para. 17 of V.23 that: *"For monitoring the emission reductions from project activities, project participants must refer to applicable provisions for monitoring plan for all project types and small-scale project activities. For PoAs, coordinating/managing entities must refer to those provisions for all project types, small-scale project activities and PoAs in the Project standard."*
- Clean Development Mechanism Project Standard for programmes of activities V.3.0 whereby it is stated that:

77. The projects participants may use sampling for the determination of parameter values for calculating GHG emission reductions if the applied methodologies, the applied standardized baselines or other methodological regulatory documents, allow this. In such cases, the project participants shall develop and describe a sampling plan in accordance with the "Standard: Sampling and surveys for CDM project activities and programme of activities".

- Guideline: Sampling and surveys for CDM project activities and PoAs V.4.0 whereby it is stated that:

29. "The sampling plan should contain information relating to: (A) sampling design; (B) data to be collected; and (C) implementation plan."

For each VPA, monitored emission reductions generated over all ICS batches (age classes) are determined by:

1. Calculating the amount of emission reductions generated by each batch of ICSs based on the monitored number of distributed ICSs, quantity of woody biomass saved per ICS type (via ICS efficiency or fuel use) and taking account of the overall proportion of operating ICSs, the residual consumption of baseline stoves and the specific size of the age class.
2. Summing up the emission reductions obtained for each age class.

Aside from the directly measured number of project devices distributed (per household) and corresponding dates of commissioning of project devices/batches, all monitored parameter values are measured through a representative sample of recorded ICSs own by *households*.

(a) Sampling Design

Objectives and reliability requirements

The sampling objective is to provide unbiased and reliable estimates of these parameter values during the crediting period with the confidence/precision level required by AMS-II.G. and summarized in the table 4 below. This would be achieved through a representative sample of distributed ICS, as compared to the study of the total population of cooking stoves distributed.

Target population and sampling frame

The *target population* is the totality of ICSs (sampling unit) distributed, among which sampling is performed to monitor $n_{y,i,j}$ and μ_y .

For the monitoring of parameter $\eta_{\text{new},i,j}$, which is sensitive to the aging of stoves, all cooking stoves sold are grouped in batches (age classes) and efficiency parameter values are estimated for each batch as a separate population, and for the other parameters sampling (in particular with regard to minimum sample size calculation) can be carried out over several batches as one population.

Stoves pertaining to the same batch is defined as all stoves that have been sold within one year/365 days meaning that first stove sold and last stove sold should have maximum 365 days difference. For instance, if the « eldest » stove of the population has been sold the first of January, the « youngest » stove should have been sold the 31 December of the same year at last. If the « eldest » stove has been sold the 1st February, the « youngest » stove that can be part of the population to be sampled should have been sold the 31st January of the following year.

The *sampling frame* is the data on ICS sales entered and/or available in the VPA's electronic database.

The electronic database records information for each sale at least on the following:

- Sales date (age class)/date of replacement of the ICS
- Project stove serial number(s)
- Type and size of ICS(s)
- Customer (and reseller as applicable) name
- Contact details of customer (and reseller as applicable)

- User type (e.g. household)¹⁷

Sampling Method

Due to the homogeneity requirement for grouping VPAs/PoAs under one sampling plan, the sampling method is simple random sampling¹⁸ for all parameters monitored through sampling at all times.

Sample Size

A sample size is calculated in order to meet reliability requirements.

The project proponent has the possibility of:

- sampling of a group of VPAs and PoAs under one sampling plan,
- sampling of each VPA individually.

Parameter values are estimated by sampling in accordance with the requirements in AMS-II.G. separately and independently for each of the VPAs included in the PoA except when a single sampling plan covering a group of VPAs and PoAs is undertaken.

In the latter case, the populations of all VPAs and PoAs in the group are combined together, the sample size is determined and a single survey is undertaken to collect data.

A pre-condition for grouping VPAs and PoAs is that they consist of the same end-users (e.g. households), which own the same type, model and size of ICS (Jiko-type Holy cook size M), i.e. the same technical specifications, located in the same boundaries. This allows to estimate parameters $n_{y,i,j}$ and μ_y as well as $\eta_{\text{new},i,j}$ without bias.

In case the project proponent(s) and/or the CME(s) decide jointly to choose grouped sampling, a single sampling plan of the group of VPAs/PoAs needs to be submitted.

The sample size for estimating the proportional parameters $n_{y,i,j}$ and μ_y (or their reverse) is calculated using the formula provided in the "Guideline: Sampling and surveys for CDM project activities and PoAs" para 12 of Appendix 1:

¹⁷ Note that target end-users, when buying an ICS will sign a sales agreement in which they agree to either replace the ceramic liner when not working anymore (in this case, the age of ICS is taken from the replacement date of ceramic liner) or to replace the ICS with a new one with comparable technical specifications. In the case that end user is not willing to replace/repair ICS, the ICS will be removed from the database and does not further claim ERs. Indeed, field agent when monitoring the operating status of stoves will check if stoves deployed are still operating thus if stoves where not operating anymore the stove is removed from the database. In the case liner is replaced during one monitoring period, the stove is considered not operating during the period for which the sampling is valid and will be considered as 'not in use'. The stove will be considered as 'in use' for the following monitoring period. E.g. through a program generating random numbers selecting sampled cooking stoves before the start of each monitoring or sales period to prepare participating cooking stove owners well in advance.

¹⁸ E.g. through a program generating random numbers selecting sampled cooking stoves before the start of each monitoring or sales period to prepare participating cooking stove owners well in advance.

$$n_p \geq \frac{t_{\alpha/2}^2 N_y \times p(1-p)}{(N_y-1) \cdot 0.1^2 \times p^2 + \frac{t_{\alpha/2}^2 p(1-p)}{2}} \quad \text{Equation (1)}$$

Where:

$t_{\alpha/2}$ Student's t Critical Values equal to 1.96 in the case when 95% confidence interval and a 10% margin of error are required and value; equal to 1.645 in the case when 90% confidence interval and a 10% margin of error are required

N_y Size of the population of stoves considered for the monitoring session
 p Expected proportion

The monitoring of $n_{y,i,j}$ and $\mu_{y,i,j}$ is based on the same sample, which is the sample with the larger sample size of the two.

If it is not possible to meet the 95/10 confidence/precision, then:

- for $\mu_{y,i,j}$, the higher bound of the 95%/10% confidence/precision requirements shall be used as the correct value.
- for $n_{y,i,j}$, the lower bound of 95%/10% confidence/precision shall be used as the correct value.

AMS-II.G. provides four options to determine $B_{y,savings,i,j}$. Option 3 (Water Boiling Test) has been chosen.

Accordingly, for $\eta_{new,i,j}$ ¹⁹ the sample size is calculated using the formula mentioned in the "Guideline: Sampling and surveys for CDM project activities and PoAs" (simple random sampling, para.51 of Appendix 1) applicable to the determination of a mean value parameter:

$$n_{\eta} \geq \frac{t_{\alpha/2}^2 N_{age,y} V}{(N_{age,y}-1) \cdot 0.1^2 + t_{\alpha/2}^2 V} \quad \text{Equation (2)}$$

Where:

$$V = \left(\frac{\text{standard deviation}}{\text{mean}} \right)^2$$

$N_{age,y}$ Number of stoves distributed belonging to the age class y
 $t_{\alpha/2}$ Student t critical values

¹⁹ For ICS other than medium-size, $\eta_{new,i,j}$ is determined for a representative sample of corresponding operative ICSs.

In general, if the sample size calculation returns a value of less than 30 samples, a minimum sample size of 30 is chosen when the parameter of interest is a proportion. If the parameter of interest is a numeric mean value (i.e. not a proportion or percentage) the student's t-distribution is used if the resulting sample size is less than 30.

(b) Data to be collected

The table below summarizes the variables to be measured and the main specifications and modalities required for data collection including confidence/precision level to assure and control for quality of the sampled data (QA/QC). Furthermore, the implementation plan below reveals additional QA/QC measures.

Table 4: Monitored parameters and specifications for monitoring

Parameter	Description	Confidence / Precision level ²⁰	Source of data (and method)	Grouped VPAs/PoAs	Frequency	Seasonality ²¹	Option selected to determine $B_{y,savings,i,j}$ in AMS-II.G.
$n_{y,i,j}$ ²²	Proportion of distributed ICS still operating in year y	95/10 (biennial) 90/10 (annual)	Monitoring (Visual inspection, field measurement)	Simple random sampling for all VPAs with the same boundary (or for each VPA)	At least once every two years (biennial)	Unlikely to be affected by seasonal influence	Option 3 selected
$\mu_{y,i}$	Proportion of pre-project stoves still in use in year y	95/10 (biennial) 90/10 (annual)	Monitoring results (Interview of end-user, field measurement)	Simple random sampling for all VPAs with the same boundary (or for each VPA)	At least once every two years (biennial)	Unlikely to be affected by seasonal influence	Option 3 selected
$\eta_{new,i,j}$	Efficiency of the device being deployed as part of the project activity in year y	90/10	Water Boiling test (laboratory)	Simple random sampling for all VPAs with the same boundary (or for each VPA)	Annually	Not affected by seasonal influence	Option 3 selected

²⁰ As per Tables 9 and 12 as well as para. 46 of AMS-II.G..

²¹ The assessment is based on a study carried out in the neighbouring Togo by the Cristo (2007) demonstrates that seasons do not affect the wood fuel consumption of users.

²² If more convenient, the project proponent may choose to monitor reverse proportions such as $1-n_{y,i,j}$ or $1-\mu_{y,i}$, as per Standard for "Sampling and surveys for CDM project activities and programme of activities" para. 11 (a). If the project proponent decides to monitor the reverse ($1-n_{y,i,j}$ and/or $1-\mu_{y,i}$) he can receive the value of $n_{y,i,j}$ and $\mu_{y,i}$ as follows: $1-(1-n_{y,i,j})$ or $1-(1-\mu_{y,i})$.

(c) Implementation Plan

The CME and management of the monitoring team coordinate the overall implementation of the monitoring and sampling plan. The monitoring team consists of one monitoring manager and one data manager. The implementation plan consists of the following principal steps:

Step 1: Selection of all monitoring options at time of VPA inclusion, in particular for monitoring $B_{y,savings,i,j}$ (option 2, 3 or 4 according to AMS-II.G.)

Step 2: Stove distribution and sales agreements

Person in charge: VPA implementer and CME

The CME cross-checks data of at least 1% of monthly signed sales agreements registered in the electronic database. According to the *Standard for "Sampling and surveys for CDM project activities and programme of activities"* this is representative²³. This step is implemented continuously throughout the implementation plan.

Step 3: Compilation of data into electronic database

Person in charge: VPA data manager

Two different persons are in charge of recording sales agreements electronically so they can control the quality of each other's work. This step is implemented continuously throughout the implementation plan.

Step 4: Choice of number of VPAs/PoAs under single sampling plan and number of monitored batches

Person in charge: VPA implementer in accordance with CME

Step 5: Sample size calculation including oversampling

Person in charge: VPA data manager

Step 6: Sample drawing

Person in charge: VPA data manager

A sample of distributed ICSs is selected randomly out of the sampling frame per batch for monitoring $n_{y,t,j}$, $\mu_{y,j}$, $B_{y=1,new,i,j,survey}$, $B_{new,KPT,i,j}$, $\eta_{new,j}$ and $SC_{new,i,j}$, respectively.

Step 7: Data collection

Person in charge: VPA data manager

The data manager organizes the monitoring campaigns, which includes lists of sampled users with contact details but without the stoves' serial numbers. This way the serial

²³ For instance, according to the Standard for "Sampling and surveys for CDM project activities and programme of activities", for a binary parameter and a sample of 15,000 ICSs with an assume proportion of 95%/5% a sample of 15 ICSs is calculated (section B.7.2). The minimum percentage is therefore $15/15,000=0.1\%$. 1% is therefore conservative.

number recorded during the campaign can be matched with records in the electronic database to assess the accuracy of the measurement. The data of sampled users' is collected through surveys, which are filled at the operation site of the stove (a priori, costumer's address). Questions to the end-user operationalize the monitored parameters. If required, the survey includes an additional question on the baseline stove to rule out the end-use of certain baseline stoves and verify the baseline of the specific VPA.

To measure the ICS efficiency $\eta_{\text{new},j}$ (and optionally $SC_{\text{new},j}$) the stove is brought to a laboratory and tested as per monitoring specification and QA/QC requirements above.

Step 8: Consolidation of monitoring results

Person in charge: VPA data manager

Step 9: Record keeping

Person in charge: VPA data manager

All monitored data is recorded in hard copy and in the electronic database. All documents, lists and questionnaires produced during monitoring are saved electronically with physical copies securely stored under the direct responsibility of the data manager and the supervision of the monitoring manager. All data is kept and archived electronically for two years after the end of the crediting period or the last issuance of certified emission reductions, whichever occurs later.

B.7.3. Other elements of monitoring plan

>> The CME is responsible for overall monitoring organization. The sampling plan, data collection & consolidation and results analysis are implemented by an adequately trained monitoring team, well aware of GS requirements and supervised by the CME. The monitoring team consists of one monitoring manager and one data manager. If external expertise or support in management or controlling is needed or to avoid conflict of interest, the CME may contract local third-party experts (e.g. NGOs) assist the monitoring team.

A compensation system, possibly based (in parts) on objective performance criteria, may be introduced in order to encourage workers.

Table 5: Monitoring team and responsibilities

Role	Responsibilities
Monitoring manager	- In general, ensure that all VPAs follow the monitoring plan - Ensure that the equipment and measurements are in line with the measurement methods, recording frequency and archiving approaches in monitoring plan - Ensure that monitoring data collected is consolidated and entered in electronic database - Ensure that monitoring team receives proper training

	- Ensure a coherent and standard monitoring report for each VPA
Data manager	- Collect monitoring data - Enter data in electronic database and archive hardcopies - Carry out sample size determination and emission reduction calculations

Before the commencement of monitoring works, the CME ensures that the monitoring staff / VPA implementers (if applicable) receive *training* according to their responsibilities in the monitoring. For all staff, it involves information on the general PoA management system so that roles, responsibilities and communication channels are clear.

Document evidence on training schedules, sessions and trainers is recorded and reflected in the monitoring reports.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of VPA

>> 12/07/2021

The starting date of the VPA has been determined as the sales date of the first cook stoves, which is after the start of PoA validation in accordance with the Glossary of CDM terms. The PoA was validated in 2014 and its crediting period has been renewed, spanning from 02/09/2020 to 01/09/2025.

C.1.2. Expected operational lifetime of VPA

The expected operational lifetime of the VPA002 is about 15 years. In the meantime, improved cook stove will be continuously monitored to ensure they are in working order, as necessarily agreed with participating households in their first ICS purchase contract and such as described in the monitoring plan.

C.2. Crediting period of project

C.2.1. Start date of crediting period

>> 12/07/2021 (1st Crediting Period).

C.2.2. Total length of crediting period

>> 5 years renewable starting from 12/07/2021 to 11/07/2026.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 1. Human rights	Man and Man Enterprise elaborated business code of conduct and Human rights statements referring explicitly to its commitments.
Principle 6. Labour rights	The project developer will provide proof documents of employment, training, payment and safety equipment for each monitoring periods. Man and Man Enterprise elaborated business code of conduct and Human rights statements referring explicitly to its commitments.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project being a cook stove project, it is not a gender sensitive project. The project does not harm neither women or men nor girls and boys. To the contrary, the superior, more efficient combustion process significantly improves the air quality, thus helping to reduce respiratory disorders specially of women and children. Furthermore, the higher efficiency helps save money for buying charcoal. Finally, the project will also reduce the proportion of time women spend or unpaid domestic work related to collection of fuel as well as cooking meals on inefficient cookstoves. The gained time can then be applied elsewhere in economic or social pursuits
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	Ghana's Ministry of Gender, Children and Social Protection (MoGCSP) was created in January, 2013 as a successor to the

	Ministry of Women and Children's Affairs.²⁴ In 2015, Ghana also approved a National Gender Policy²⁵, which, among others, aims at promoting a "the application of sustainable energy sources particularly for women and households" and a "saving culture among women, especially women in the informal sector." (p.29 & p. 30)
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No, the project will be ensuring that women will increase their access to, or control of benefits yielded by the project, specifically around fuel collection and cooking, and hence an expert is not required for this.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No, as the project activity is increasing the role of women through its application, and women have formed a major part of the local stakeholder consultation (visible on photos of the LSC, questions, attendance sheets etc.), and the UN does not discriminate on gender for returnees receiving aid packages (and hence stoves), an expert would not be required

²⁴ <https://www.mogcsp.gov.gh/about/>

²⁵ <https://www.mogcsp.gov.gh/mdocs-posts/national-gender-policy/>

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

On the 9th of December 2021, the Local Stakeholder Consultation of the VPA took place in the Longji Hotel Tarkwa, Ghana. 26 people participated in the meeting ; the project was presented, questions were raised and answered. The overall feedback was very positive as reflected in the evaluation forms. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	Located at: Santasi New Site, near Kwadee Junction Plot 13 B, Kumasi, Ghana
GS Contact (mandatory)	help@goldstandard.org
Other	Contact person: Michael Yaw Agyei Tel.: +233 (0) 243 47 36 42 manandman.ent@gmail.com

SECTION F. Eligibility and inclusion criteria for VPAs inclusion

To be included in the PoA, the VPA proposed should verify all applicability assessments below:

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
1	Project Area, Project Boundary and Scale /Location of project	The SSC-VPA takes place within the borders of Ghana (a host country which has ratified the Kyoto protocol and is listed as a Non-Annex I country with no cap GHG emissions); Only Carbon Dioxide (CO₂), Methane (CH₄) and/or Nitrous Oxide (N₂O) are eligible for Gold Standard crediting, provided Projects comply with Gold Standard Requirements and eligibility criteria.	Mapped locations of stoves distribution area provided / The SSC-VPA boundaries are the centres (urban, peri-urban and rural) of Western region in Ghana which are within borders of Ghana such as demonstrated by the GPS coordinates in section A.2.²⁶ Cross-check with registries of other standards revealed SSC-VPA is not included in other standards. Methodology AMS II.G V.12.0 applied through the PoA-DD is limited to reductions in CO₂. Ghana does not have an emission cap in place at the moment of validation of this VPA-DD
2	Specification of the technology/measure	The SSC-VPA specifies the level and type of service provided by the technology/measure as well as its performance which are in line with the technology outlined in SSC-PoA-DD and in compliance with national and/or international testing/certification requirements. Specifications of the technology/measure shall include the type, capacity and other key features of the design of the systems;	Certified ICS testing report of medium size ICS is provided. / The SSC-VPA involves the marketing of improved cook stoves, with a certified efficiency of 30.9% which is at least 20% such as described in section A.3. ICS deployed are Jiko type system technology composed of two parts, a metallic part and a ceramic part that retain the heat and make it more efficient than traditional system. The capacity (power output of the system is 0.95 KW). ICS deployed are portable.
3	Target group	The target group of each SSC-VPA shall be either:	Section A.1. Project description in VPA-DD

²⁶ The sale agreement of each ICS sold under the VPA will be collected with information about address in order to check that each ICS is sold to users in Ghana but cannot be provided at the time of VPA inclusion thus it will be reflected in the monitoring report.

- The households (Households are domestic unit consisting of the members of a family who live together and share food and cooking equipment).
 - The communities (communities are groups of individuals that have an identity and share some of their resources such as biomass and cooking equipment. For instance, schools, police camps, prisons are considered as communities)
 - The small and medium enterprises (Small and Medium Enterprises (SME) are small and medium business that use stoves as equipment for their business and consume biomass in order to sell meals (commercial use). For instance, restaurant is considered as a SME as they use ICSs to cook and sell meals.).
 - Charcoal stove users and/or three-stone fire users.
- And Project Design study and/or any references that can substantiate the compliance. / The SSC-VPA involves the marketing of ICS for charcoal using households of urban, periurban and rural centers of Ghana as described in section A.3.

4

AMS.II-G version 12.0

Compliance with the applicability

The baseline consumption and of the applied baseline stove efficiency of methodology target group selected should be determined.

FAO. 2017. The charcoal transition: greening the charcoal value chain to mitigate climate change and improve local livelihoods, by J. van Dam. Rome, Food and Agriculture Organization of the United Nations

The SSC-VPA involves the marketing of ICS for charcoal using households of urban, rural, periurban and rural centers of Ghana as described in section A.5.

/ The SSC-VPA involves the marketing of ICS for *households*, which are consuming in the baseline 5.64 tonnes of wood-fuel

equivalent per year *in the absence of the project activity*.²⁷

The efficiency of baseline stoves is determined at 20%

5	Compliance with the applicability of the applied methodologies / Type of project	The project shall be the distribution (sale) of efficient cook stoves to target user in Ghana, and therefore classifies as an End-use Energy Efficiency Improvement project.	Section A.5. Project description in VPA-DD / The SSC-VPA involves the sales of ICS such as stated in section A.5 and thus demonstrates that it classifies as an End-use Energy Efficiency Improvement project. Since there are GS approved CSA requirements, the project type is automatically eligible (cf. para. 4.1.3 of GS4GG Principles and requirements.) End-use energy efficiency such as efficient cooking is a pre-identified CSA project type, as per para. 3.1.1 of GS4GG CSA requirements V.1.2.
6	Demonstration of additionality	In reference to the "Annex B (Positive List) of the CSA Requirements, V. 1.2", the annual energy savings generated by each ICS sold under the framework of a VPA included under this PoA shall not exceed 600 MWh, which is equivalent to 1,800 MWh _{th} (microscale CDM unit);	Detailed calculations are provided in Section A.4.: Application of methodology(ies) in VPA-DD And ICS test report. / The SSC-VPA nominal energy savings of each ICS Nominal energy saving of each ICS <i>medium size</i> is 8.62 MWh _{th} /y which is lower than 600 MWh, which is equivalent to 1,800 MWh _{th}
7	Compliance with the applicability of the applied methodologies.	The VPA complies and applies the small-scale baseline and monitoring methodology AMS II.G Version 12.0;	Section B.2.: Application of methodology(ies) in VPA-DD. And The SSC-VPA properly applies methodology AMS-II.G Version 12.0 in compliance with its eligibility criteria as follow: AMS II.G version 12.0 comprises the introduction of improved cookstoves which is typically the case of the proposed SSC-VPA.

²⁷ Baseline consumption may be updated along project implementation if more recent data become available.

		It is proven that non-renewable biomass has been used and forest area has reduced since 1990.
8	Transfer of credits ownership shall be guaranteed from stove user to VPA implementer. Other requirements of the applied methodologies The VPA is either implemented by the Coordinating/managing entity or by another entity which has signed a binding agreement with the CME which ensures that they are aware and agree that their activity is subject to a PoA and that their carbon rights have to be relinquished to the CME;	Contractual agreement between CME and VPA implementer (necessary only if CME is not the VPA implementer). And / or Sales agreement between VPA implementer and end user. / The SSC-VPA is implemented by Man and Man Enterprise which is also the CME of the PoA thus no contractual agreement is needed. Each end user included into the program voluntarily adheres to its terms and conditions, including a waiver for the transfer of credit ownership to the project owner, Man and Man Enterprise, either through individual sales agreements or retailers' charter.
9	Double counting Each cook stove sold under the SSC-VPA will be uniquely marked using ICS serial number and recorded in an electronic database for all PoA life, as described in SSC-PoA-DD section B in order to ensure that no double counting will occur ;	Section C of the SSC-PoA and template of sale contract in which serial number of ICS that will be distributed under the PoA appears and show serial number structure ensuring that ICS to be allocated with a unique serial no to ensure no double counting will occur. / As stated in section C of PoA-DD and section A.13 of VPA-DD, each ICS sold under the SSC-VPA will be identified based on the unique serial number riveted under the stove and recorded in order to avoid double counting intra PoA (misallocation or double counting within the different VPAs of the PoA) and double counting inter-PoA (misallocation between two different PoAs or related to cook stoves not belonging to the PoA).
10	Unregistered/deregistered activity The SSC-VPA-DD has been reviewed by the CME and submitted to a VVB for inclusion into the PoA; Among others, the CME shall check that VPAs are neither registered as project activities, included in another registered	Conclusion review of the SSC-VPA-DD of the CME. And/or Quotes of VVB signed for VPA inclusion. / A letter from the CME to the VVB (cc/ VPA implementer) submitting the proposed VPA-DD for inclusion into the PoA is

		PoAs, nor the project activities that have been deregistered;	proving that the SSC-VPA has been verified by CME and submitted to a VVB for review before inclusion in the PoA.
11	Local stakeholder consultation and environmental impact analysis	A local stakeholder consultation has been conducted prior to the inclusion of the SSC-VPA in the PoA The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	Section F of PoA-DD or section E. of the VPA-DD, LSC report submission notification and consultation minutes and presentation. / A local stakeholder consultation has been conducted at the VPA level for household's end-users in urban areas. Details pertaining to the consultation are available in LSC report.
12	Sampling	The SSC-VPA sampling plan complies with the "<i>Standard for sampling and surveys for CDM project activities and programme of activities</i>" V.09.0;	Section B.7.2 of SSC-VPA-DD and / or additional supporting evidence if applicable. / As stated in the monitoring plan outlined in the PoA-DD, the sampling effort aims to achieve for the annual inspection the level of accuracy (95/10 if cross-VPA monitoring is chosen or 90/10 if standalone VPA monitoring is chosen) which is compliant with AMS-II.G. version 12.0 and the "<i>Standard for sampling and surveys for CDM project activities and programme of activities</i>".
13	Diversion of official development assistance	The SSC-VPA does not involve any public funding according to the OECD definitions for Official Development Assistance (ODA);	ODA Declaration (as per GHG Emissions Reduction & Sequestration Product Requirements) / As stated in A.7, the project does not involve any public funding according to the OECD definitions for Official Development Assistance (ODA).
14	Start date	The starting date of the SSC-VPA is verifiable through documentary evidence and shall not be prior the start date of the PoA.	Document evidence such as order of first batch of ICSs or first sale agreement signed. / As stated in section C.1.1, the starting date of the SSC-VPA has been determined as the sale of the first ICS which is verifiable the first sale agreement signed by household which should not be before the start date of PoA.
15	Gold Standard requirements (SDG and Safeguarding)	The SSC-VPA shall respect the requirements of the Gold Standard regarding SDG and Safeguarding Principles assessments.	VPA-DD and related evidence / The VPA describes in section the Local Stakeholder Consultation Report and sections B and D of the VPA-DD how it will respect the requirements of the Gold

Principles assessments)		Standard regarding SDG and Safeguarding Principles assessments, as well as monitoring of SDG parameters.
16 Local stakeholder consultation and environmental impact analysis	An Environmental Impact Assessment (EIA) of the project shall be conducted if required by the national laws.	A local consultation took place in Tarkwa on the 9 th December, 2021, in order to share information with the local residents/entities of interest, and obtain feedback from them. More details available on the SCR./ Since no Environmental Impact Assessment (EIA) is required by the national laws for VPA002 2, no EIA report is provided.
17 Gold Standard requirements (Host Country Requirements)	Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	This project is in compliance with Ghana's legal, environmental, and social regulations. The project is actively contributing to the Host Country by reducing the GHG emissions, fuel consumption as well as economic and time savings to thousands of poor households. The company is registered and has legal right to operate. Therefore no other permission/permit is required. M&M also received report from EPA regarding environmental impact of stoves production.
18 Gold Standard requirements (Contact details)	The Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors).	GS Cover letter Extract from the trade registries and documentation by the governing jurisdiction / Cf. Appendix 1 of VPA-DD and GS cover letters provided. Extract from the trade registries and documentation by the governing jurisdiction shall be provided.
19 Gold Standard Requirements (Legal Ownership)	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated.	Sample of sale agreements / Cf. section A.3 of VPA-DD The proofs that end-users are aware of and willing to give up their rights on Products shall be provided.

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	The programme protects human rights including freedoms, cultural property and no discrimination policy. Man and Man Enterprise elaborated business code of conduct and Human rights statements referring explicitly to its commitments. Ghana has ratified the Universal Declaration of Human Rights and the African Charter on Humans and People's Rights.	-
Principle 2. Gender Equality			

1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No	Improved cookstoves manufacturing, distribution and use by Man and Man Enterprise is a gender-friendly activity since it results in positive impacts on women and children who benefit the most from cleaner cooking and lower fuel consumption/spendings. Besides, the projects does neither directly nor indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women, and applies the principles of non-discrimination, equal treatment, and equal pay for equal work, both on customers' end (retailing & after-sales) and employees' end (hiring & training). Last but not least, Ghana has adopted a National Gender Policy (Ministry of Gender, Children and Social Protection, 2015) for "mainstreaming gender equality and women's empowerment into Ghana's development efforts.	-
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Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	Project proponent trains people to manage ICS production resulting in increasing qualifications of Ghanaians people. The project factory improves working conditions and safety measures for craftsmen as compared to baseline informal workshops.	-
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project does not include cultural and historic sites. None of the manufacturing, distributions sites have a particular historical, traditional or religious value.	
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No		
>>			
Principle 4.2 Forced Eviction and Displacement			

Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project require neither physical or economic relocation of peoples.	
>>			
Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	The risk or expected issue is not relevant to the Project.	-
b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?			
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?		The project is not operating in area claimed by indigenous people.	
>>			

Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	As ratified by Ghana under the United Nations Convention Against Corruption, it is explicitly referred under article XI of Man and Man Enterprise's business code of conduct and Human rights statement that employees may not bribe anyone for any reason, whether in dealings with governments or the private sector.	-
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be		The project adheres to all labour laws and requirements.	

documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and			
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reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	The project activity consists in retailing improved cookstoves that sell at a margin and reduce households’ fuel spendings drastically, with payback period of months only.	-
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The project is expected to reduce GHG emissions by 191,627 tCO ₂ /year on average.	-
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			

Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. "Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?"	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 9.2 Vulnerability to Natural Disaster			

Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or	No	The risk or expected issue is not relevant to the Project.	-

use of hazardous and non-hazardous chemicals and/or materials?			
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The risk or expected issue is not relevant to the Project.	-
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The risk or expected issue is not relevant to the Project.	-
>>			

Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?		The project decreases the consumption of firewood having a positive impact on conserving forest ecosystems	
>>			
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?		No	
>>			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	Man and Man Enterprise
Registration number with relevant authority	BN - 1,698D
Street/P.O. Box	Kumasi Santasi New site-Kumasi
Building	-
City	Kumasi
State/Region	Ashanti region
Postcode	Ks 4588
Country	Ghana
Telephone	+233 (0) 243 47 36 42
E-mail	-
Website	manandman.ent@gmail.com
Contact person	Mr. Agyei Michael Yaw
Title	
Salutation	Mister
Last name	Yaw
Middle name	Michael
First name	Agyei
Department	-
Mobile	-
Direct tel.	+233 (0) 243 47 36 42
Personal e-mail	manandman.ent@gmail.com

APPENDIX 3- LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	
Risk of change to the Project activities during Project Certification Period:	
Land-use history and current status of Project Area:	
Socio-Economic history:	
Forest management applied (past and future)	
Forest characteristics (including main tree species planted)	
Main social impacts (risks and benefits)	
Main environmental impacts (risks and benefits)	
Financial structure	
Infrastructure (roads/houses etc):	
Water bodies:	
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	
Where indigenous people and local communities are situated:	
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	

APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Revision History

Version	Date	Remarks
2.0	4 May 2022	
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption